COMPUTER ENGINEERING

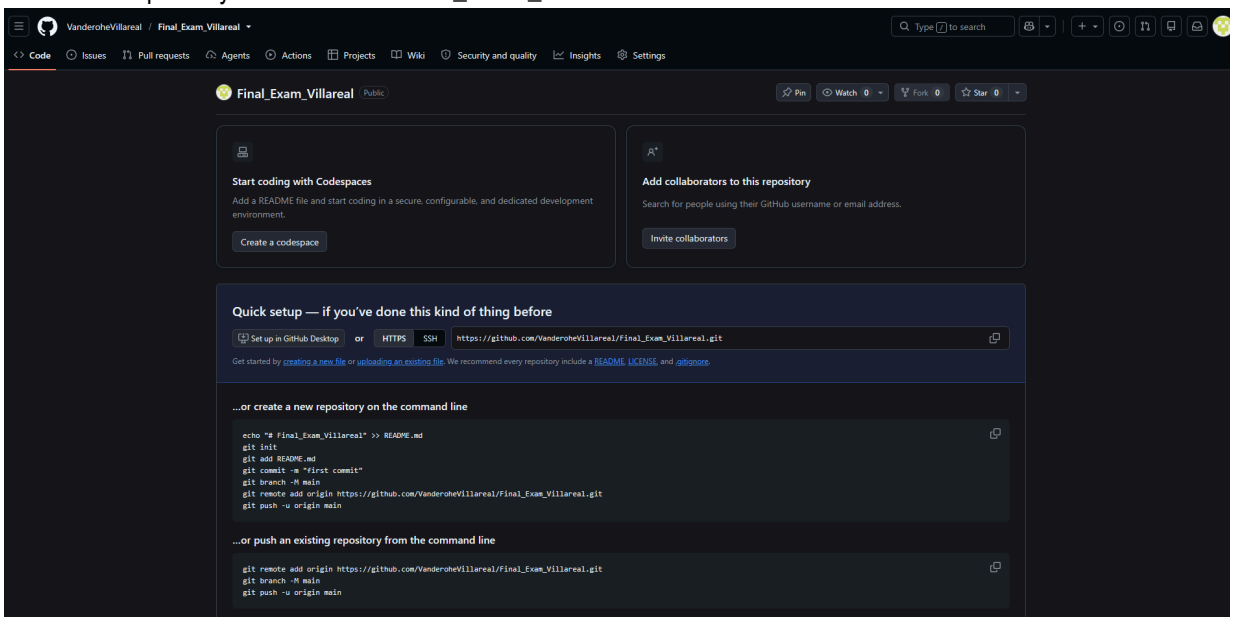
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COURSE CODE/TITLE:	CPE 212 | Automating Server Management	**SECTION:**	CPE31S2
DATE PERFORMED:	10/29/26	**DATE SUBMITTED:**	10/29/26
NAME:	Villareal, Vanderohe Vernon	**INSTRUCTOR:**	Engr. Jimlord Quejado

FINALS **Final Hands-on Skills Exam**

1. PROCEDURE OUTPUT

1. Create a repository and label it as "Final_Exam_Surname"



2. Clone your new repository in your VM



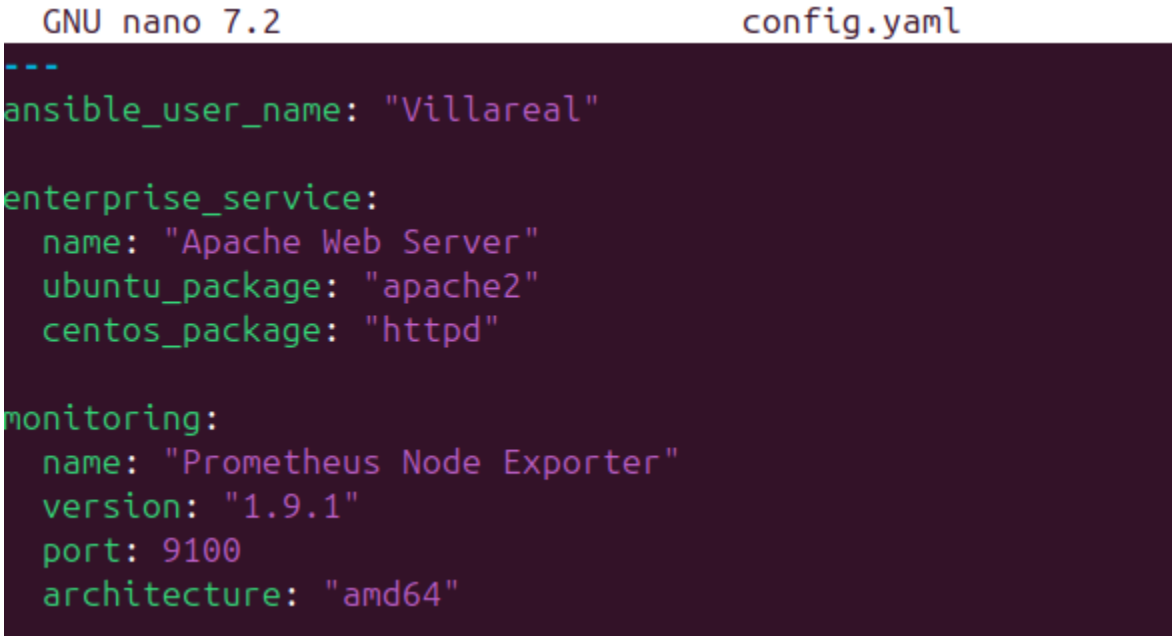


3. Create an Ansible playbook that does the following with an input of a config.yaml file and structure inventory file.

OS	IP address
Ubuntu	*192.168.56.139*
CentOS	*192.168.56.137*

3.1 Install and configure one enterprise service that can be installed in Debian and Centos servers

Configuration File (config.yaml)



Main ansible playbook (site.yaml file)

GNU nano 7.2 site.yaml

```yaml
---
- name: Configure Ubuntu and CentOS Servers
  hosts: all
  become: true
  vars_files:
    - config.yaml

  tasks:
    - name: Update Ubuntu package cache
      ansible.builtin.apt:
        update_cache: true
      when: ansible_os_family == "Debian"

    - name: Install Apache on Ubuntu
      ansible.builtin.apt:
        name: apache2
        state: present
      when: ansible_os_family == "Debian"

    - name: Install Apache on CentOS
```

GNU nano 7.2 site.yaml

```yaml
      ansible.builtin.dnf:
        name: httpd
        state: present
      when: ansible_os_family == "RedHat"

    - name: Start and enable Apache on Ubuntu
      ansible.builtin.service:
        name: apache2
        state: started
        enabled: true
      when: ansible_os_family == "Debian"

    - name: Start and enable Apache on CentOS
      ansible.builtin.service:
        name: httpd
        state: started
        enabled: true
      when: ansible_os_family == "RedHat"

    - name: Download Node Exporter
```

3.2 Install and configure one monitoring tool that can be installed in Debian and Centos servers (if it is a stack there should be option of different host)

```
GNU nano 7.2                                site.yaml
      url: "https://github.com/prometheus/node_exporter/releases/download/v{{>
      dest: /tmp/node_exporter.tar.gz
      mode: "0644"

  - name: Extract Node Exporter
    ansible.builtin.unarchive:
      src: /tmp/node_exporter.tar.gz
      dest: /tmp
      remote_src: true
      creates: "/tmp/node_exporter-{{ monitoring.version }}.linux-{{ monitori>

  - name: Install Node Exporter binary
    ansible.builtin.copy:
      src: "/tmp/node_exporter-{{ monitoring.version }}.linux-{{ monitoring.a>
      dest: /usr/local/bin/node_exporter
      remote_src: true
      owner: root
      group: root
      mode: "0755"
```

```
GNU nano 7.2                                site.yaml
    ansible.builtin.copy:
      dest: /etc/systemd/system/node_exporter.service
      owner: root
      group: root
      mode: "0644"
      content: |
        [Unit]
        Description=Prometheus Node Exporter
        After=network.target

        [Service]
        Type=simple
        ExecStart=/usr/local/bin/node_exporter --web.listen-address=:{{ monit>
        Restart=on-failure

        [Install]
        WantedBy=multi-user.target

  - name: Start and enable Node Exporter
    ansible.builtin.systemd:
```

After running the playbook

```
PLAY RECAP *********************************************************************
centos-node                : ok=9    changed=8    unreachable=0    failed=0    s
kipped=3    rescued=0    ignored=0
ubuntu-control             : ok=9    changed=1    unreachable=0    failed=0    s
kipped=3    rescued=0    ignored=0
```

4.4 Change Motd as "Ansible Managed by <username>"

```
- name: Configure MOTD
  ansible.builtin.copy:
    dest: /etc/motd
    content: "Ansible Managed by {{ ansible_user_name }}\n"
    owner: root
    group: root
    mode: "0644"
```

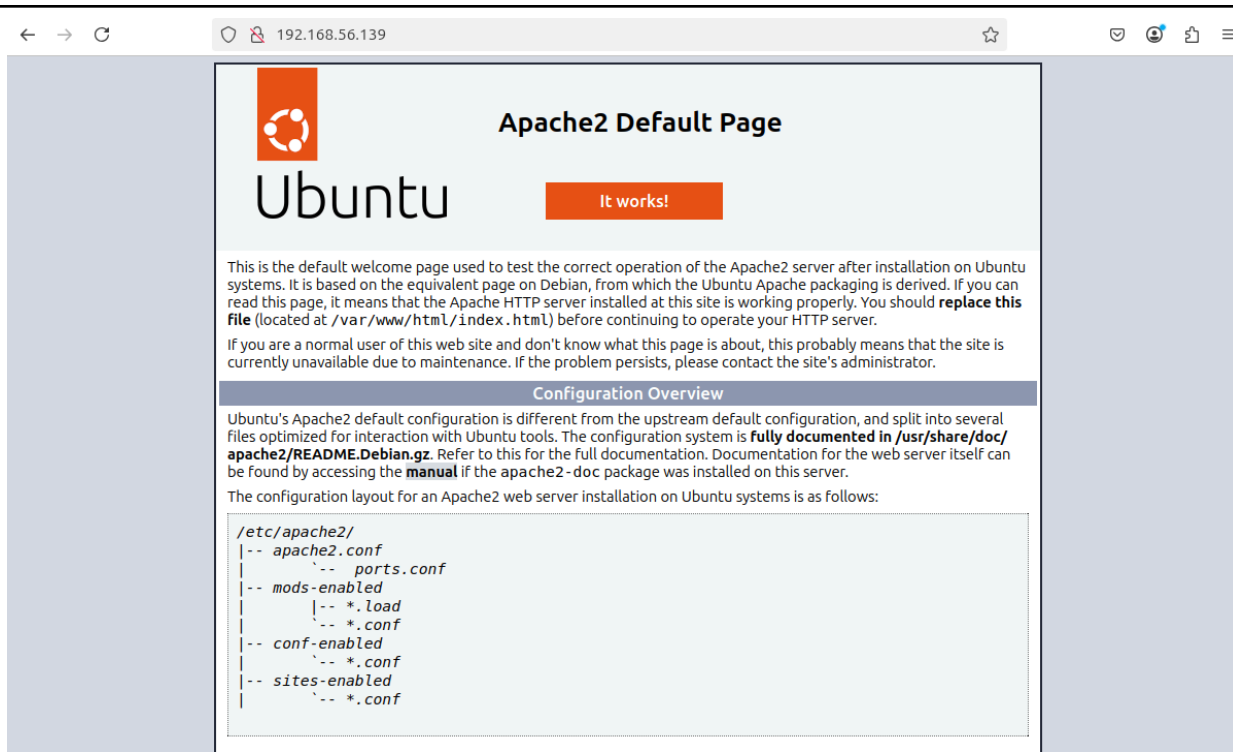
```
villareal1@workstation:~/Final_Skills_Exam$ cat /etc/moto
Ansible Managed by Villareal
```

Verification for the installation

UBUNTU

```
illareal1@workstation:~/Final_Skills_Exam$ systemctl status apache2
● apache2.service - The Apache HTTP Server
     Loaded: loaded (/usr/lib/systemd/system/apache2.service; enabled; preset:
     Active: active (running) since Thu 2026-10-29 01:12:07 UTC; 37min ago
       Docs: https://httpd.apache.org/docs/2.4/
   Main PID: 11404 (apache2)
      Tasks: 6 (limit: 3457)
     Memory: 11.3M (peak: 11.6M)
        CPU: 369ms
     CGroup: /system.slice/apache2.service
             ├─11404 /usr/sbin/apache2 -k start
             ├─11407 /usr/sbin/apache2 -k start
             ├─11408 /usr/sbin/apache2 -k start
             ├─11409 /usr/sbin/apache2 -k start
             ├─11410 /usr/sbin/apache2 -k start
             └─11411 /usr/sbin/apache2 -k start
```

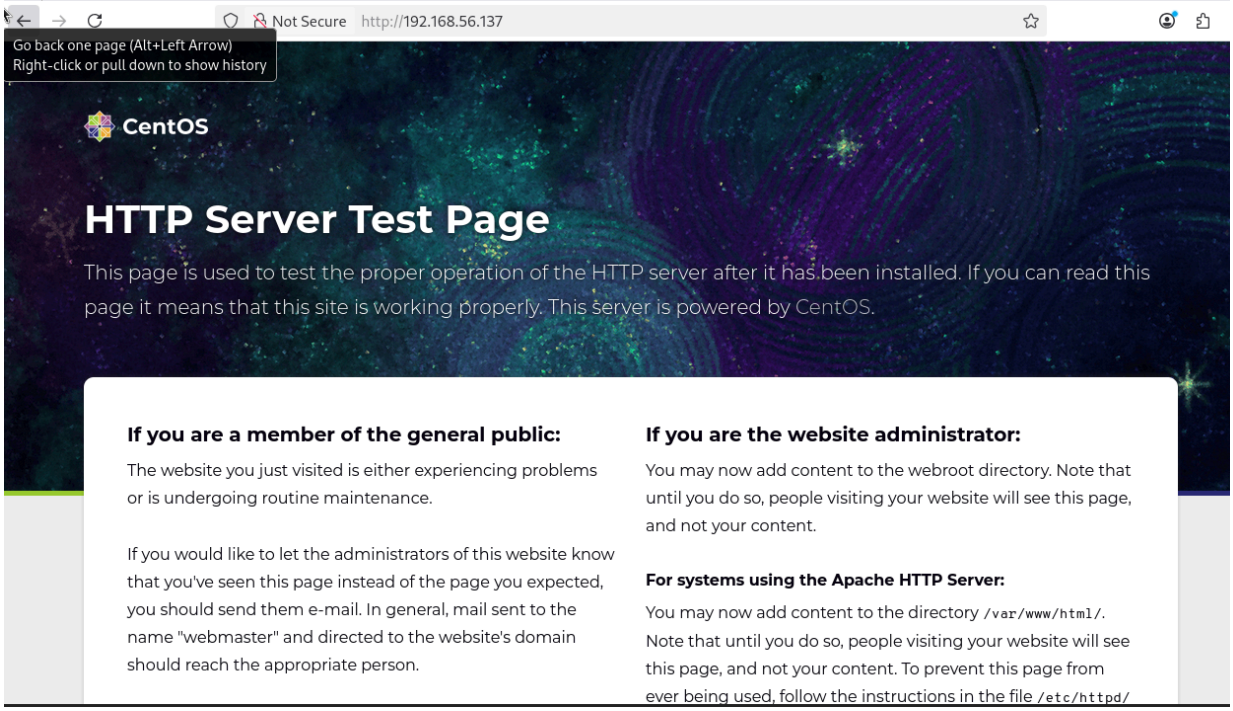
```
villareal1@workstation:~/Final_Skills_Exam$ systemctl status node_exporter
● node_exporter.service - Prometheus Node Exporter
     Loaded: loaded (/etc/systemd/system/node_exporter.service; enabled; preset
     Active: active (running) since Thu 2026-10-29 01:42:18 UTC; 12min ago
   Main PID: 22589 (node_exporter)
      Tasks: 6 (limit: 3457)
     Memory: 2.3M (peak: 3.0M)
        CPU: 37ms
     CGroup: /system.slice/node_exporter.service
             └─22589 /usr/local/bin/node_exporter --web.listen-address=:9100
```



CENTOS

```
[villareal@localhost ~]$ systemctl status httpd
● httpd.service - The Apache HTTP Server
     Loaded: loaded (/usr/lib/systemd/system/httpd.service; enabled; preset: di
     Active: active (running) since Thu 2026-10-29 09:46:55 PST; 4min 23s ago
       Docs: man:httpd.service(8)
   Main PID: 8920 (httpd)
     Status: "Total requests: 0; Idle/Busy workers 100/0;Requests/sec: 0; Bytes
      Tasks: 177 (limit: 22964)
     Memory: 14.7M (peak: 15.0M)
        CPU: 580ms
     CGroup: /system.slice/httpd.service
             ├─8920 /usr/sbin/httpd -DFOREGROUND
             ├─8984 /usr/sbin/httpd -DFOREGROUND
             ├─8996 /usr/sbin/httpd -DFOREGROUND
             ├─8998 /usr/sbin/httpd -DFOREGROUND
             └─9061 /usr/sbin/httpd -DFOREGROUND
```

```
[villareal@localhost ~]$ systemctl status node_exporter
● node_exporter.service - Prometheus Node Exporter
     Loaded: loaded (/etc/systemd/system/node_exporter.service; enabled; preset
     Active: active (running) since Thu 2026-10-29 09:47:58 PST; 7min ago
   Main PID: 22336 (node_exporter)
      Tasks: 5 (limit: 22964)
     Memory: 5.1M (peak: 5.6M)
        CPU: 60ms
     CGroup: /system.slice/node_exporter.service
             └─22336 /usr/local/bin/node_exporter --web.listen-address=:9100
```



4. Push and commit your files in GitHub

5. Make sure to show evidence of input (codes) process (codes successfully running) and output (evidence of installation)

5. For your final exam to be counted, please paste your repository link as an answer in this exam.

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

GITHUB LINK:
https://github.com/VanderoheVillareal/Final_Exam_Villareal

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: **None**

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: **None**

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

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Extent of Use: **None**

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

By submitting this activity, I affirm that my use of AI complies with T.I.P.'s AI usage policy and does not exceed the permitted limits for similarity, automated content generation, or academic assistance.

I understand that any false, incomplete, or misleading disclosure may be subject to investigation in accordance with due process and may result in sanctions for violation of existing T.I.P. policies.

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